BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

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Data Analytics Internship

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Executive Summary

The mutual fund industry generates vast amounts of financial and transactional data that can provide valuable insights into fund performance, investor behaviour, portfolio diversification and market trends. However, extracting meaningful information from raw datasets remains a significant challenge due to the complexity and volume of available data. The objective of this project was to develop a comprehensive Mutual Fund Analytics Platform capable of transforming financial data into actionable business intelligence through data engineering, analytical modelling and visualisation techniques.

The project involved the collection and integration of multiple mutual fund datasets, including Net Asset Value (NAV) history, Assets Under Management (AUM), investor transaction records, benchmark indices, portfolio holdings and scheme performance metrics. These datasets were processed through a structured ETL pipeline to ensure data quality, consistency and reliability before analysis.

Using Python, Pandas, SQLite and SQL, extensive data preparation activities were carried out including data cleaning, validation, transformation and consolidation. Exploratory Data Analysis (EDA) was subsequently performed to identify industry trends, investor behaviour patterns and fund performance characteristics. The insights generated during this phase formed the basis for dashboard development and advanced financial analytics.

Interactive Power BI dashboards were designed to provide a consolidated view of industry performance, investor activity, risk-return characteristics and market trends. These dashboards enable users to explore data through interactive filters and visualisations while supporting informed decision-making.

In addition to descriptive analytics, advanced analytical models were implemented to assess investment risk and investor behaviour. These included Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio Analysis, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration Analysis using the Herfindahl-Hirschman Index (HHI), and a rule-based Fund Recommendation Engine.

The project successfully demonstrates how data analytics, financial modelling and business intelligence tools can be integrated to generate actionable insights within the mutual fund industry. The resulting platform provides a scalable framework for evaluating fund performance, monitoring investor behaviour and supporting data-driven investment decisions.

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1. Introduction

The mutual fund sector represents one of the most important segments of the financial services industry. Mutual funds enable investors to participate in diversified portfolios managed by professional fund managers while reducing the complexity associated with direct investment in financial markets. As investor participation continues to increase, Asset Management Companies (AMCs) must analyse growing volumes of data to understand performance trends, investor preferences and portfolio risks.

Traditionally, financial data is distributed across multiple systems and datasets, making it difficult to generate a unified view of performance and risk. This challenge creates a strong need for integrated analytical platforms capable of consolidating data, performing advanced analysis and presenting findings in a format that supports strategic decision-making.

The Bluestock Mutual Fund Analytics Platform was developed to address this requirement. The project combines data engineering, financial analytics and business intelligence techniques to provide a complete analytical framework for mutual fund analysis. By integrating data from multiple sources, the platform enables comprehensive evaluation of industry performance, investor behaviour, portfolio diversification and investment risk.

The project follows a structured workflow beginning with data acquisition and ETL processing. Once the data was cleaned and standardised, exploratory analysis was performed to identify key patterns and trends. Financial performance metrics were subsequently calculated, and advanced risk analytics techniques were applied to evaluate downside risk, diversification and investor retention. Finally, interactive Power BI dashboards were developed to visualise findings and support decision-making.

The project demonstrates practical applications of modern data analytics techniques within the financial domain while highlighting the value of combining statistical analysis, database management and business intelligence tools to generate meaningful insights.

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2. Project Objectives

The primary objective of this project was to design and develop a comprehensive analytics platform capable of providing insights into mutual fund performance, investor behaviour and market trends.

The specific objectives of the project were as follows:

- To collect, clean and integrate mutual fund datasets from multiple sources.
- To develop a structured ETL pipeline for reliable data processing and storage.
- To analyse industry growth trends through Assets Under Management (AUM), folio counts and SIP inflows.
- To evaluate mutual fund performance using financial metrics such as CAGR, Sharpe Ratio, Alpha, Beta and Maximum Drawdown.
- To understand investor behaviour through demographic, geographic and transaction-based analysis.
- To identify downside risk using Value at Risk (VaR) and Conditional Value at Risk (CVaR).
- To measure portfolio concentration and diversification using the Herfindahl-Hirschman Index (HHI).

• To identify investor retention risks through SIP continuity analysis.

• To develop a rule-based recommendation engine for fund selection based on risk preferences.

• To create interactive dashboards for visual exploration and business decision support.

By achieving these objectives, the project aims to demonstrate how analytical techniques can be applied to improve transparency, risk management and decision-making within the mutual fund ecosystem.

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3. Data Sources

The effectiveness of any analytical platform depends heavily on the quality and diversity of its data sources. To ensure comprehensive analysis, multiple datasets covering different aspects of the mutual fund industry were utilised throughout this project.

The primary datasets included:

NAV History Data

This dataset contains historical Net Asset Value information for mutual fund schemes. NAV data forms the foundation for return calculations, risk analysis and performance evaluation.

Fund Master Data

The fund master dataset contains scheme-level information including fund names, categories, fund houses and investment plans. This dataset serves as the central reference table used throughout the project.

Investor Transaction Data

Investor transaction records provide details regarding purchases, redemptions and SIP contributions. These records were used to analyse investor behaviour, transaction patterns and demographic trends.

Assets Under Management (AUM) Data

AUM data was utilised to assess industry growth and identify leading Asset Management Companies based on total managed assets.

Benchmark Index Data

Benchmark data, including NIFTY indices, was used to compare mutual fund performance against broader market movements.

Portfolio Holdings Data

Portfolio holdings information provides details regarding sector allocations and portfolio composition. This dataset was used to calculate portfolio concentration and diversification metrics.

Scheme Performance Data

This dataset contains performance-related indicators including returns, volatility, Sharpe Ratio, Alpha, Beta and expense ratios.

Monthly SIP Data

Monthly SIP datasets were utilised to analyse systematic investment behaviour and identify long-term contribution trends.

The integration of these datasets enabled the development of a comprehensive analytical framework capable of evaluating multiple dimensions of mutual fund performance and investor activity.

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4. ETL Design and Architecture

A robust ETL (Extract, Transform and Load) framework was developed to ensure data quality, consistency and accessibility throughout the analytical process.

The ETL architecture consisted of four primary stages:

Extraction

The extraction stage involved collecting data from multiple CSV files containing mutual fund information, investor transactions, portfolio holdings, benchmark indices and industry statistics. These datasets originated from different sources and required consolidation before analysis.

Transformation

The transformation stage focused on improving data quality and preparing datasets for analysis. Key transformation activities included:

- Removal of duplicate records.

- Handling of missing values.

- Standardisation of date formats.

- Data type conversion.

- Validation of numerical fields.

- Consistent naming conventions across datasets.

- Creation of derived analytical variables.

These processes ensured that the datasets were suitable for downstream analytical tasks.

Loading

Following transformation, cleaned datasets were stored within a centralised SQLite database. This approach enabled efficient querying, simplified data management and supported integration across analytical workflows.

The SQLite database served as the single source of truth for all subsequent analysis and dashboard development activities.

Analytics Layer

Once loaded into the database, analytical models were applied to generate insights. These included descriptive analytics, financial performance metrics, risk models and investor behaviour analysis.

The analytics layer generated outputs that were subsequently visualised through Power BI dashboards and incorporated into advanced analytical models.

ETL Workflow

Raw CSV Files

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Data Cleaning & Validation

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Transformation & Standardisation

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SQLite Database Storage

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Financial Analytics

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Power BI Dashboard Development

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Business Insights & Recommendations

The ETL architecture ensured that all analyses were performed on reliable, standardised and validated data, thereby improving the accuracy and credibility of project findings.

5. Exploratory Data Analysis (EDA) Findings

Exploratory Data Analysis (EDA) was performed to gain an initial understanding of the mutual fund ecosystem and identify meaningful trends, patterns and relationships within the available datasets. This stage served as a critical foundation for subsequent dashboard development and advanced analytical modelling.

The primary objectives of EDA were to understand industry growth patterns, analyse investor behaviour, evaluate transaction activity and identify differences between fund categories. Through visualisation and statistical analysis, key insights were extracted from the data to support evidence-based decision-making.

The EDA process focused on the following areas:

• Industry Assets Under Management (AUM) growth

• SIP inflow trends

• Investor demographics

• Geographic investment distribution

• Fund category performance

• Transaction behaviour patterns

• Market benchmark comparisons

The analysis revealed significant growth across the mutual fund industry during the study period, accompanied by increasing investor participation and sustained SIP activity. Furthermore, differences between fund categories highlighted varying risk-return characteristics and investor preferences.

The findings generated through EDA directly influenced the design of dashboard visualisations and advanced analytical models developed later in the project.

5.1 Industry Growth Analysis

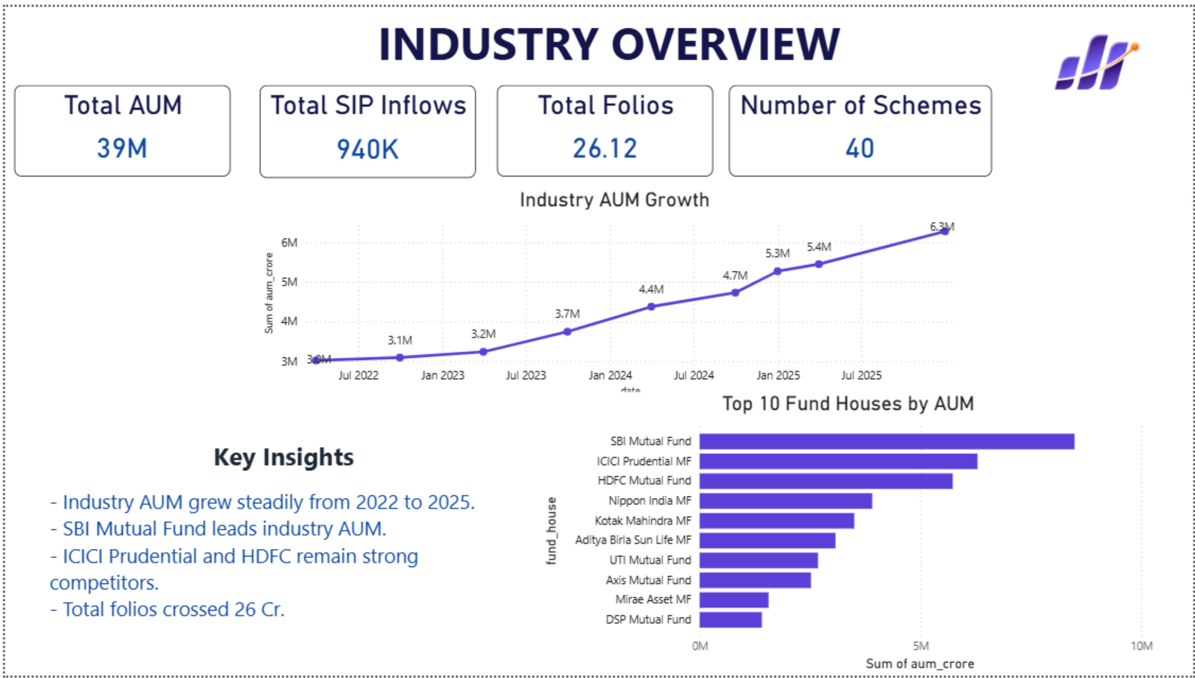

Figure 1: Industry Overview Dashboard

The Industry Overview Dashboard provides a consolidated view of key industry-level metrics including Assets Under Management (AUM), SIP inflows, folio counts and scheme distribution.

Analysis of the dashboard indicates that the mutual fund industry experienced consistent growth throughout the observation period. Total industry AUM increased from approximately 3.0 million units to 6.3 million units, representing substantial growth in managed assets. This trend reflects increasing investor confidence and expanding participation within mutual fund products.

SIP inflows remained robust throughout the period, reaching approximately 940 thousand units. The steady growth of SIP contributions demonstrates a growing preference for disciplined long-term investment strategies among retail investors.

The analysis further reveals that the total number of investor folios exceeded 26.12 crore, indicating broad participation across the market. The growing folio count suggests continued expansion of the mutual fund investor base.

Among Asset Management Companies, SBI Mutual Fund emerged as the market leader in terms of Assets Under Management. ICICI Prudential Mutual Fund and HDFC Mutual Fund also maintained significant market positions, demonstrating strong competitive presence within the industry.

Key Findings:

• Industry AUM more than doubled during the analysis period.

• SIP contributions remained consistently strong.

• Investor participation increased significantly.

• SBI Mutual Fund maintained leadership in managed assets.

• The industry exhibited stable long-term growth characteristics.

5.2 Investor Behaviour Analysis

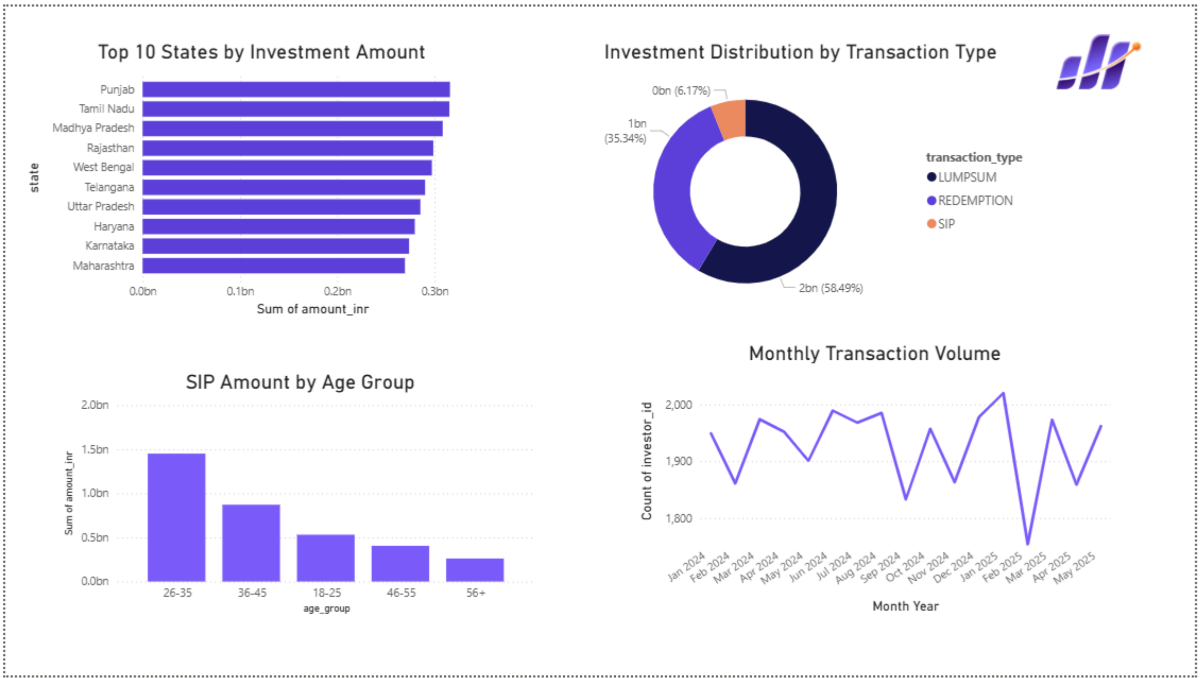

Figure 2: Investor Analytics Dashboard

Investor behaviour analysis was conducted to understand demographic patterns, geographic investment trends and transaction preferences. This analysis provides valuable insight into how different investor segments interact with mutual fund products.

State-level analysis revealed that Punjab, Tamil Nadu and Madhya Pradesh recorded the highest investment volumes. This suggests stronger participation from investors within these regions compared with other states included in the dataset.

Transaction type analysis highlighted the dominance of lumpsum investments, which accounted for the largest share of transaction volume. Redemption transactions represented a substantial portion of activity, while SIP contributions formed a smaller but steadily growing segment.

Age-group analysis identified investors aged between 26 and 35 years as the largest contributors to SIP investments. This finding aligns with expectations, as investors within this age group are typically in the early stages of wealth creation and retirement planning.

Monthly transaction volume remained relatively stable throughout the observation period. Although minor fluctuations were observed, transaction activity demonstrated overall consistency, indicating sustained investor engagement.

Key Findings:

• Punjab and Tamil Nadu recorded strong investment activity.

• Lumpsum investments dominated transaction volume.

• Investors aged 26–35 represented the most active SIP contributors.

• Monthly transaction behaviour remained stable over time.

• Investor participation was distributed across multiple demographic groups.

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6. Performance Analysis

Performance analysis was conducted to evaluate mutual fund returns, risk exposure and benchmark performance. This section focuses on understanding how different funds performed under varying market conditions while assessing the relationship between risk and reward.

Multiple financial performance indicators were considered, including return percentage, standard deviation, Sharpe Ratio, Alpha, Beta and expense ratio. These metrics collectively provide a comprehensive view of fund performance and investment efficiency.

6.1 Risk versus Return Analysis

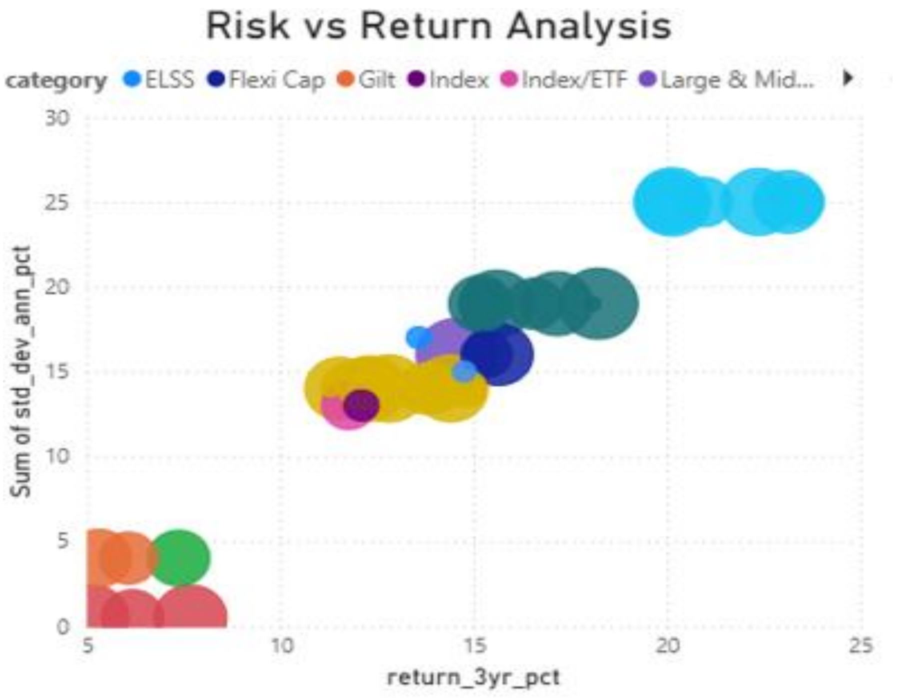


Figure 3: Risk versus Return Analysis

The Risk versus Return scatter plot illustrates the relationship between expected returns and volatility across different mutual fund categories.

A positive relationship between risk and return is evident within the dataset. Funds delivering higher returns generally exhibited higher levels of volatility, highlighting the fundamental trade-off between risk and reward. Large-cap and flexi-cap funds occupied the medium-to-high return region while maintaining manageable levels of risk.

Lower-risk categories produced more stable performance but generated comparatively lower returns. This behaviour aligns with traditional portfolio theory, which suggests that investors must accept greater uncertainty in pursuit of higher returns.

The clustering of fund categories also provides valuable insight into investment behaviour and category-specific performance characteristics.

Key Findings:

• Higher returns were generally associated with increased volatility.

• Large-cap and flexi-cap funds achieved favourable risk-return balances.

• Lower-risk funds generated more stable but modest returns.

• Category clustering reflected distinct investment strategies.

6.2 NAV versus Benchmark Analysis

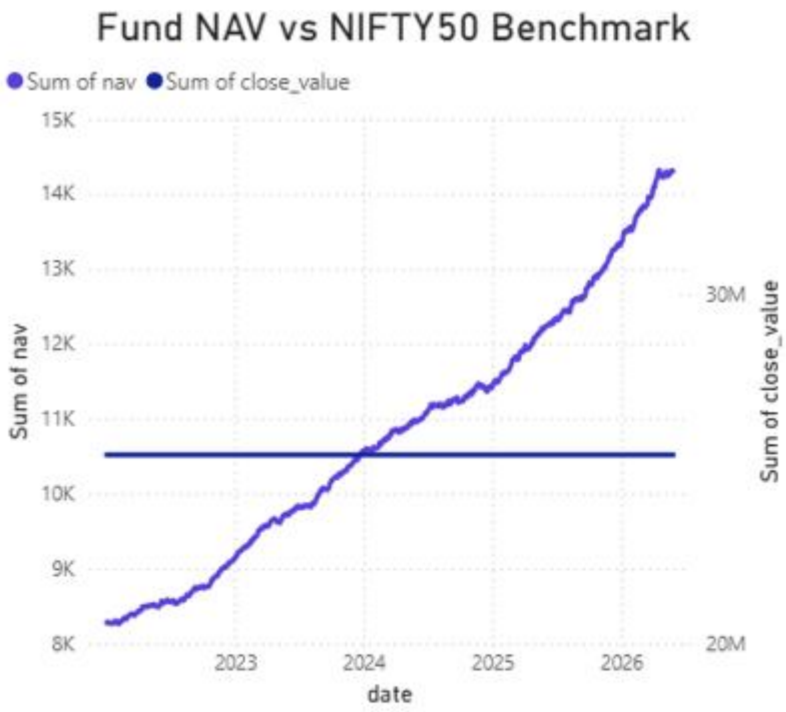


Figure 4: Fund NAV versus NIFTY50 Benchmark

Benchmark analysis was performed to compare mutual fund performance against broader market movements represented by the NIFTY50 index.

The chart indicates that fund NAV demonstrated strong long-term growth throughout the observation period. While benchmark performance remained relatively stable, fund NAV displayed substantial appreciation over time, particularly during the later stages of the analysis period.

This trend suggests that the selected fund portfolio was able to generate meaningful value for investors while maintaining favourable performance relative to benchmark conditions.

Key Findings:

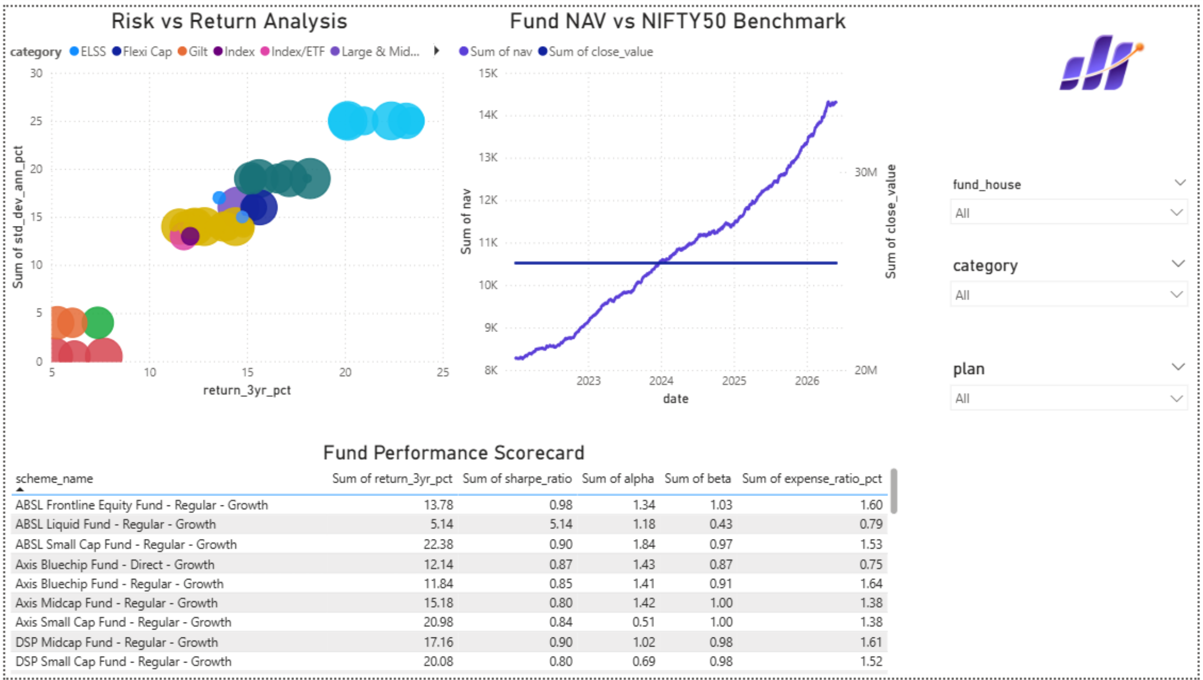
• Fund NAV demonstrated strong long-term growth.

• Performance accelerated during later periods.

• Benchmark comparison indicates positive investment outcomes.

• Long-term investment discipline produced favourable results.

6.3 Fund Performance Scorecard



scheme_name	Sum of return_3yr_pct	Sum of sharpe_ratio	Sum of alpha	Sum of beta	Sum of expense_ratio_pct
ABSL Frontline Equity Fund - Regular - Growth	13.78	0.98	1.34	1.03	1.60
ABSL Liquid Fund - Regular - Growth	5.14	5.14	1.18	0.43	0.79
ABSL Small Cap Fund - Regular - Growth	22.38	0.90	1.84	0.97	1.53
Axis Bluechip Fund - Direct - Growth	12.14	0.87	1.43	0.87	0.75
Axis Bluechip Fund - Regular - Growth	11.84	0.85	1.41	0.91	1.64
Axis Midcap Fund - Regular - Growth	15.18	0.80	1.42	1.00	1.38
Axis Small Cap Fund - Regular - Growth	20.98	0.84	0.51	1.00	1.38
DSP Midcap Fund - Regular - Growth	17.16	0.90	1.02	0.98	1.61
DSP Small Cap Fund - Regular - Growth	20.08	0.80	0.69	0.98	1.52

Figure 5: Fund Performance Scorecard

The Fund Performance Scorecard consolidates multiple financial performance indicators into a single analytical view. Metrics including returns, Sharpe Ratio, Alpha, Beta and expense ratio enable direct comparison between schemes.

The scorecard demonstrates that performance varied considerably across funds. Some schemes generated superior returns but carried higher volatility, while others delivered more stable performance with lower risk exposure.

Alpha values indicated varying levels of benchmark outperformance, while Beta values reflected differing sensitivities to market fluctuations. Expense ratios remained relatively controlled across most schemes, suggesting efficient fund management practices.

The scorecard provides investors and analysts with a practical framework for evaluating investment alternatives using multiple performance dimensions simultaneously.

Key Findings:

• Fund performance varied significantly across schemes.

• Sharpe Ratios highlighted differences in risk-adjusted returns.

• Alpha values reflected varying benchmark outperformance.

• Expense ratios remained relatively competitive.

• Multi-metric analysis improves investment evaluation accuracy.

7. Advanced Analytics

Beyond traditional descriptive analytics, advanced analytical techniques were implemented to evaluate investment risk, investor behaviour, portfolio diversification and fund suitability. These models provide deeper insights into financial performance and support data-driven investment decisions.

The advanced analytics phase focused on risk measurement, investor retention analysis, portfolio concentration assessment and personalised fund recommendation.

7.1 Value at Risk (VaR) Analysis

	amfi_code	var_95	cvar_95
0	100016	-0.014364	-0.018060
1	100025	-0.003793	-0.004994
2	100033	-0.019034	-0.023456
3	101206	-0.013282	-0.017439
4	101207	-0.026021	-0.032459

Figure 6: Value at Risk and Conditional Value at Risk Results

Value at Risk (VaR) is a widely used risk management metric that estimates the maximum expected loss within a specified confidence interval. In this project, a 95% confidence level was applied using historical return distributions.

The analysis was conducted by calculating the 5th percentile of daily return distributions for each mutual fund. This approach provides an estimate of the loss that is unlikely to be exceeded under normal market conditions.

The results indicated varying levels of downside risk across different funds. Fund 101207 recorded one of the highest downside risk values with a VaR of approximately -2.60%, suggesting greater vulnerability to adverse market conditions compared with other funds in the sample.

VaR serves as a valuable risk indicator by helping investors understand potential losses before making investment decisions.

Key Findings:

• Downside risk varied significantly across funds.

• Certain schemes exhibited relatively high loss potential during adverse periods.

• VaR provides a practical framework for evaluating investment risk.

7.2 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also referred to as Expected Shortfall, extends the VaR concept by estimating the average loss beyond the VaR threshold.

While VaR identifies the risk boundary, CVaR provides insight into the severity of losses once that boundary has been breached. Consequently, CVaR is considered a more comprehensive measure of tail risk.

The analysis revealed that Fund 101207 recorded a CVaR value of approximately -3.25%, indicating that losses beyond the VaR threshold could become significantly larger during extreme market conditions.

CVaR enables investors and portfolio managers to better understand the magnitude of worst-case scenarios and improve risk management strategies.

Key Findings:

• CVaR provides deeper insight into extreme downside risk.

• Tail-risk exposure differed across mutual funds.

• Certain schemes demonstrated significantly larger losses beyond VaR thresholds.

7.3 Rolling 90-Day Sharpe Ratio Analysis

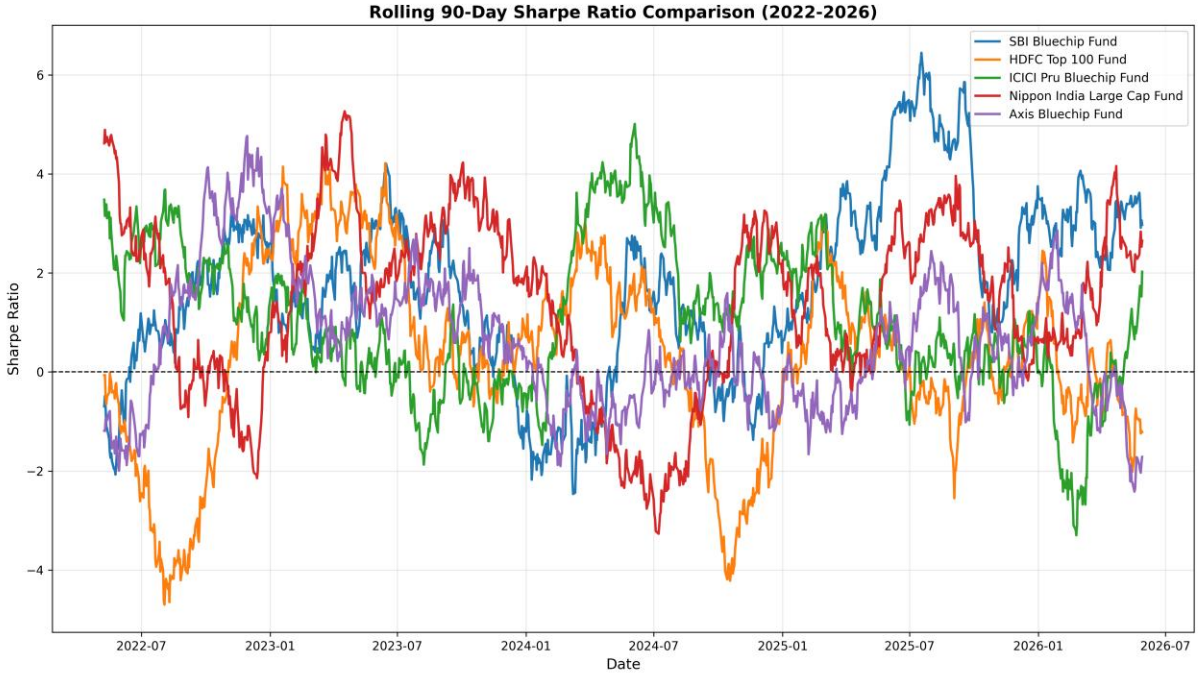


Figure 7: Rolling 90-Day Sharpe Ratio Analysis

The Rolling Sharpe Ratio was calculated using a 90-day rolling window to evaluate risk-adjusted performance over time.

Unlike static performance metrics, rolling analysis captures changes in performance across different market environments. This approach allows analysts to observe periods of strength and weakness while identifying shifts in investment efficiency.

The analysis included selected large-cap mutual funds such as SBI Bluechip Fund, HDFC Top 100 Fund, ICICI Prudential Bluechip Fund, Axis Bluechip Fund and Nippon India Large Cap Fund.

Results demonstrated varying patterns of risk-adjusted performance throughout the observation period. SBI Bluechip displayed strong Sharpe Ratio peaks during several periods, while other funds experienced greater fluctuations in performance.

Key Findings:

- Risk-adjusted performance changed significantly over time.
- SBI Bluechip exhibited strong Sharpe Ratio peaks.
- Performance leadership shifted between funds across market cycles.
- Rolling analysis provided deeper insight than static performance metrics.

7.4 Investor Cohort Analysis

	first_year	investors	average_sip	total_invested	favorite_fund
0	2024	4803	107422.54	3491125187	Mirae Asset Emerging Bluechip Fund - Regular - ...
1	2025	197	109158.58	30455243	SBI Small Cap Fund - Direct Plan - Growth

Figure 8: Investor Cohort Analysis

Investor Cohort Analysis was conducted by grouping investors according to the year of their first investment activity.

This approach enables organisations to evaluate long-term investor behaviour, identify valuable customer segments and measure retention performance across cohorts.

The analysis revealed that the 2024 cohort represented the largest investor group, consisting of approximately 4,803 investors. This cohort also generated the highest investment volume, contributing more than ₹349 crore in total investments.

Average SIP contributions remained strong across cohorts, indicating sustained investor engagement following acquisition.

Key Findings:

• The 2024 cohort was the largest investor segment.

• Total investments exceeded ₹349 crore.

• Cohort analysis provides valuable customer lifecycle insights.

• New investor acquisition remained strong during 2024.

7.5 SIP Continuity Analysis

	investor_id	sip_count	max_gap_days	status
0	INV000001	6	265.0	AT_RISK
1	INV000008	6	165.0	AT_RISK
2	INV000010	6	139.0	AT_RISK
3	INV000011	7	125.0	AT_RISK
4	INV000012	8	132.0	AT_RISK

Figure 9: SIP Continuity Analysis

SIP Continuity Analysis was performed to identify investors at risk of discontinuing systematic investment plans.

The analysis measured the gap between consecutive SIP transactions and classified investors as AT_RISK whenever the maximum gap exceeded 35 days.

Results indicated that a significant number of investors experienced irregular contribution patterns. These investors were flagged for potential churn risk and may benefit from targeted engagement initiatives.

Monitoring SIP continuity allows fund houses to improve investor retention and strengthen long-term investment participation.

Key Findings:

• Multiple investors exhibited SIP gaps exceeding 35 days.

• At-risk investors were successfully identified.

• SIP continuity monitoring supports retention strategies.

• Early intervention may reduce investor churn.

7.6 Portfolio Concentration Analysis (HHI)

	amfi_code	hhi_index	portfolio_type
0	119551	0.1187	Highly Diversified
1	119552	0.1080	Highly Diversified
2	119598	0.1073	Highly Diversified
3	119599	0.1748	Moderately Diversified
4	100016	0.1395	Highly Diversified

Figure 10: HHI Portfolio Concentration Analysis

Portfolio diversification was evaluated using the Herfindahl-Hirschman Index (HHI).

The HHI measures portfolio concentration by summing the squared portfolio weights of holdings. Lower HHI values indicate higher diversification, while larger values suggest greater concentration risk.

The results demonstrated that most mutual fund portfolios maintained HHI values below 0.15, indicating strong diversification characteristics. Only a small number of portfolios exhibited moderate concentration levels.

These findings suggest that the majority of funds effectively distribute investments across multiple holdings, thereby reducing concentration risk.

Key Findings:

• Most funds exhibited strong diversification.

• HHI values remained below concentration thresholds.

• Concentration risk was generally low.

• Portfolio diversification contributed to risk reduction.

7.7 Fund Recommendation Engine

A rule-based recommendation engine was developed to demonstrate practical application of financial analytics for investment decision support.

The recommendation system categorised investors into three risk profiles:

Low Risk Investors:
Recommended funds with the lowest volatility.

Moderate Risk Investors:
Recommended funds with balanced volatility and return characteristics.

High Risk Investors:
Recommended funds with the highest Sharpe Ratios.

The model successfully generated personalised fund recommendations based on historical risk and performance metrics. While the system uses a rule-based approach, it demonstrates the potential application of analytics-driven investment advisory solutions.

Key Findings:

• Risk-based recommendations were successfully generated.

• Historical volatility and Sharpe Ratios provided effective selection criteria.

• The model can support investment decision-making processes.

• Future enhancements may incorporate machine learning techniques.

8. Dashboard Screenshots and Visualisation Analysis

The Power BI dashboards developed during this project serve as the primary interface for exploring mutual fund performance, investor behaviour, market trends and risk metrics. The dashboards were designed to provide stakeholders with intuitive access to key performance indicators while enabling interactive exploration through filters, drill-through functionality and dynamic visualisations.

The dashboard suite consists of four dedicated analytical pages covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. Together, these dashboards provide a comprehensive view of the mutual fund ecosystem.

8.1 Industry Overview Dashboard

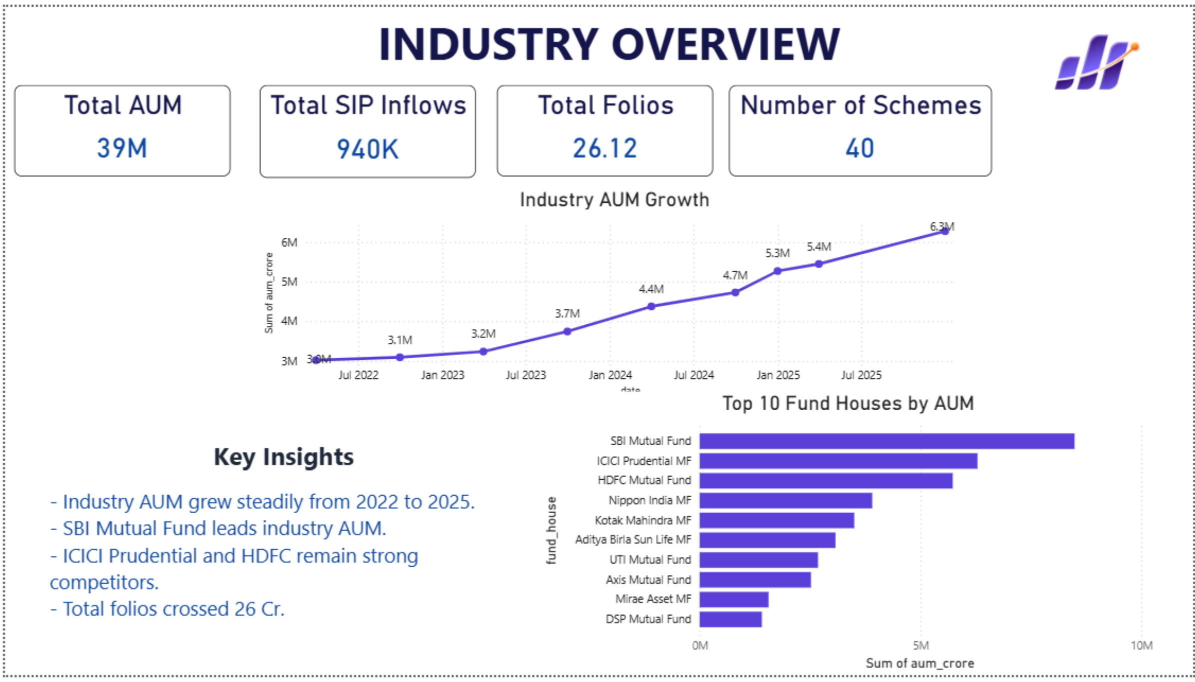


Figure 11: Industry Overview Dashboard

8.2 Fund Performance Dashboard

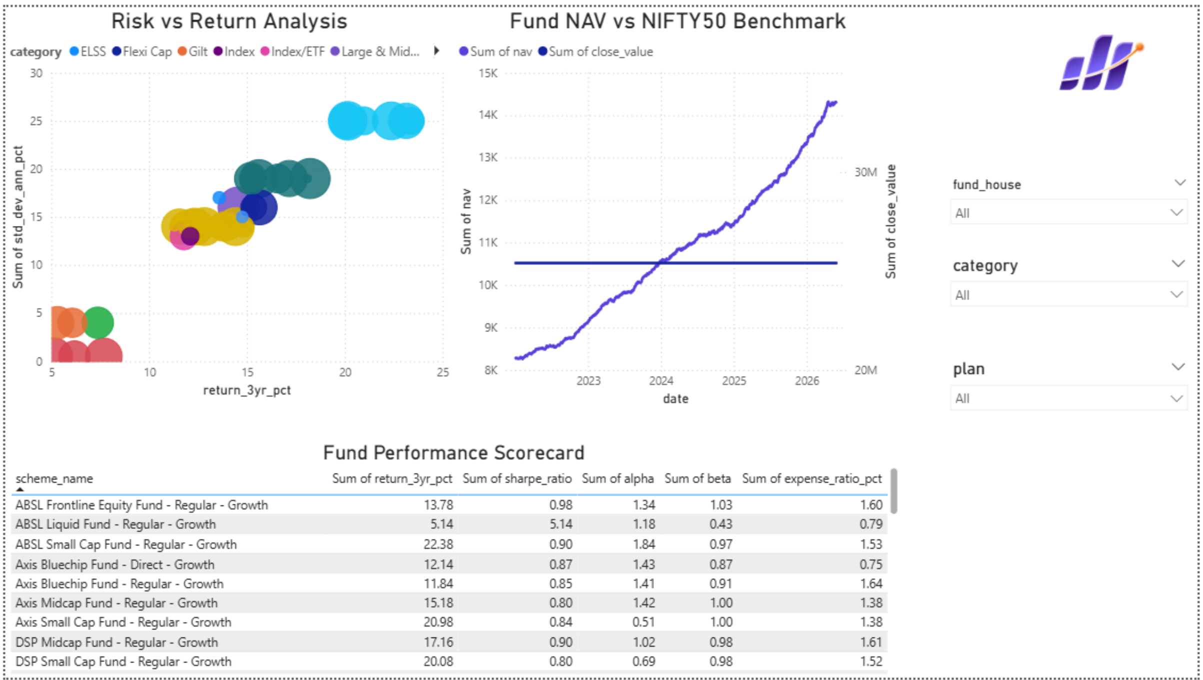


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Figure 12: Fund Performance Dashboard

8.3 Investor Analytics Dashboard

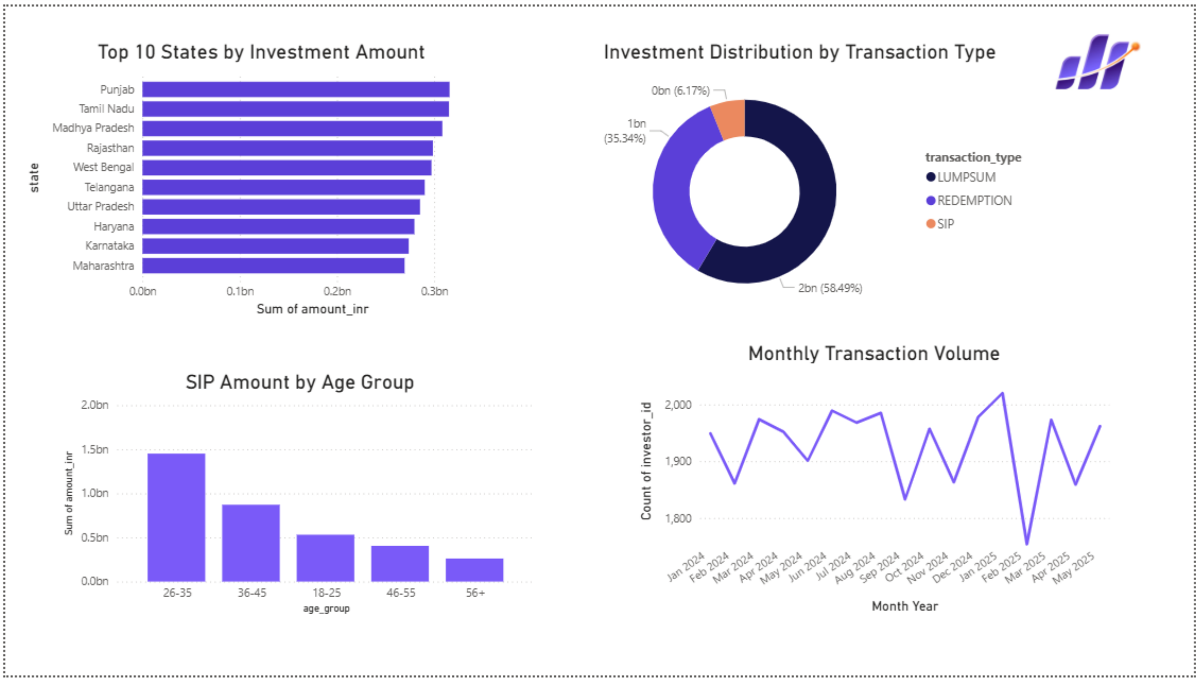

Figure 13: Investor Analytics Dashboard

8.4 SIP and Market Trends Dashboard

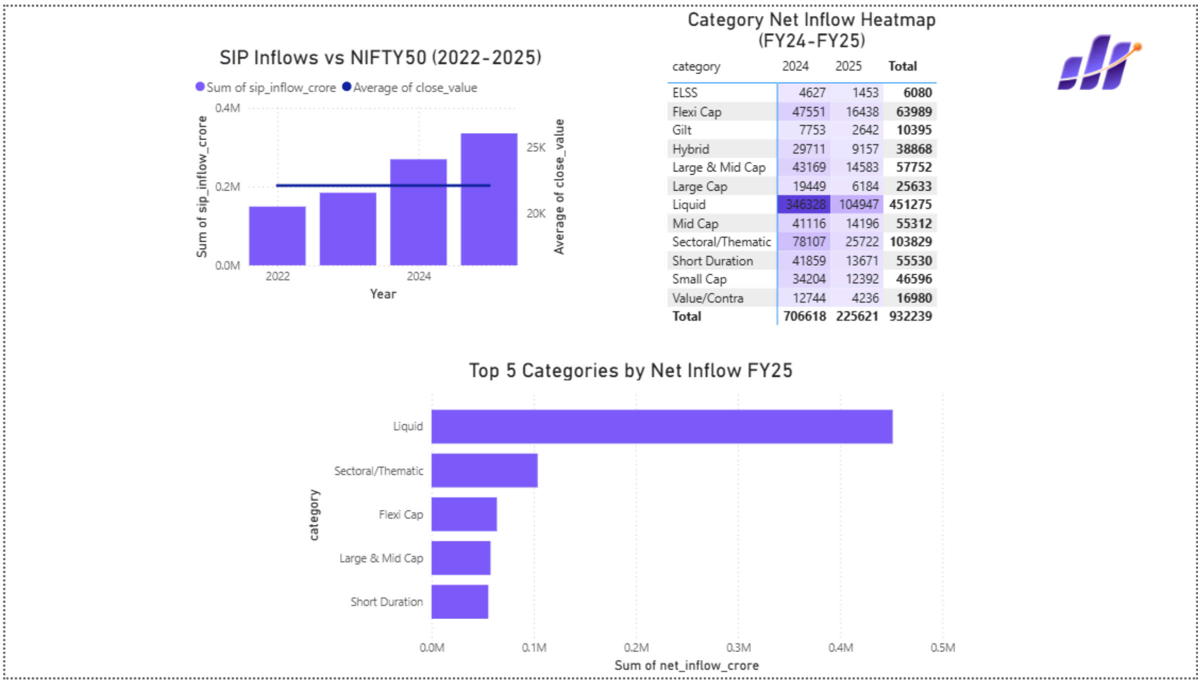

Figure 14: SIP and Market Trends Dashboard

9. Limitations

Although the project successfully demonstrates the practical application of analytics within the mutual fund industry, several limitations should be acknowledged.

Firstly, the analysis relies entirely on historical datasets. Past performance does not guarantee future results, and market conditions may change significantly beyond the scope of this study.

Secondly, the recommendation engine utilises a rule-based methodology rather than machine learning algorithms. Consequently, recommendations are generated using predefined thresholds and may not fully capture complex investor preferences or behavioural patterns.

Thirdly, the project was limited to the datasets available during the internship period. Additional information such as macroeconomic indicators, real-time market feeds, investor sentiment data and global economic variables could further enhance analytical accuracy.

Furthermore, some fund categories contained fewer observations than others, potentially affecting category-level comparisons and statistical significance.

Finally, advanced risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR) provide estimates of downside risk based on historical distributions. These techniques may not fully capture extreme market events or unexpected economic shocks.

Despite these limitations, the analytical framework developed within this project remains effective for evaluating mutual fund performance, investor behaviour and portfolio risk.

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10. Recommendations

Based on the findings generated throughout this project, several recommendations are proposed for investors, fund managers and Asset Management Companies.

Firstly, organisations should strengthen SIP retention programmes by targeting investors identified through SIP Continuity Analysis. Early intervention strategies may reduce investor churn and improve long-term engagement.

Secondly, portfolio diversification should remain a priority for both investors and fund managers. The HHI analysis demonstrated that diversified portfolios generally exhibit lower concentration risk and greater resilience during periods of market uncertainty.

Thirdly, risk management frameworks should incorporate VaR and CVaR monitoring on a regular basis. Funds exhibiting elevated downside-risk metrics should be reviewed carefully, particularly during volatile market conditions.

Fourthly, investor cohort analysis should be utilised to support customer acquisition and retention strategies. The strong performance of the 2024 cohort suggests opportunities for improving investor engagement through targeted campaigns.

Fifthly, the recommendation engine may be expanded through the incorporation of machine learning algorithms capable of generating personalised recommendations based on investor profiles, transaction history and market conditions.

Finally, dashboard adoption should be encouraged across business teams to support data-driven decision-making and improve transparency across investment operations.

11. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully integrated data engineering, financial analytics, business intelligence and risk modelling techniques into a unified analytical framework.

The project demonstrated the complete analytics lifecycle, beginning with data acquisition and ETL processing, progressing through exploratory data analysis and dashboard development, and culminating in advanced analytical models for risk assessment and investment decision support.

The Power BI dashboards developed throughout the project provided valuable insight into industry growth, fund performance, investor behaviour and market trends. Advanced analytical models including Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio Analysis, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration Analysis and the Fund Recommendation Engine further enhanced the depth and practical value of the solution.

The findings indicate strong growth within the mutual fund industry, increasing investor participation and generally well-diversified portfolios. Furthermore, the project highlights the importance of analytics-driven decision-making in supporting investment evaluation, risk management and customer engagement.

Overall, the Bluestock Mutual Fund Analytics Platform demonstrates how modern analytical techniques can transform raw financial datasets into actionable business intelligence. The resulting framework provides a scalable foundation for future enhancements and offers practical value to investors, analysts and Asset Management Companies seeking to make informed decisions within an increasingly data-driven financial environment.